# $\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$

One number computed from paired observations. Its sign is read in three ways; its size on its own is not the message.

SPLIT BY sign of $\mathrm{Cov}(X, Y)$

## CASE 01 — $\mathrm{Cov} > 0$

### Deviations move the same way

higher-than-average $X$ tends to come with higher-than-average $Y$, and lower with lower

**+** the mean product of deviations is positive

products $(X - \mu_X)(Y - \mu_Y)$ are mostly $> 0$

$X$ $Y$

EXAMPLE

height and weight across people

## CASE 02 — $\mathrm{Cov} < 0$

### Deviations move opposite ways

higher-than-average $X$ tends to come with lower-than-average $Y$

**−** the mean product of deviations is negative

products $(X - \mu_X)(Y - \mu_Y)$ are mostly $< 0$

$X$ $Y$

EXAMPLE

driving speed and trip time

## CASE 03 — $\mathrm{Cov} = 0$

### No consistent linear relationship

the deviations show no steady direction; this does **not** mean the variables are independent

**0** the mean product of deviations is zero

the products cancel out

EXAMPLE

$Y = X^2$ with $X$ symmetric about $0$: $Y$ is fixed by $X$, yet the covariance is $0$